

Particle PhysicsNotes

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January 20, 2026

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Lecture 3

1.1 Language of Particle Physics

- Theoretical/Mathematical $\rightarrow$ QFT $\rightarrow$ Symmetries
- Numerical/Computation
- Data Handling ¹

1.2 Units of Particle Physics

- Quantum Mechanics in SI² $\rightarrow$ lots of $\hbar$ and c
- adapt a unit system where $\hbar = c = 1$
This takes care of not carrying these extra factors

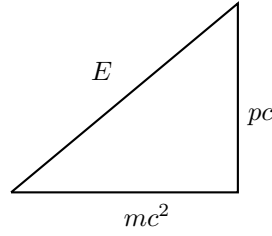
1.2.1 Outcomes of assuming this

- Velocity in special relativity $\beta = \frac{v}{c}$ becomes dimensionless
- Action is dimensionless $\rightarrow$ Energy $\times$ Time
- Mass dimension is same as energy dimension

$$[\text{mass}] = [\text{Energy}] = [\text{momentum}] \xrightarrow{\text{in}} \text{GeV}$$

$$E^2 = p^2 c^2 = m^2 c^4 \rightarrow E^2 = p^2 + m^2$$

¹in LHC large data 1 Trillion bytes per second is generated
²Standard Units

**Fig. 4.1**

- $[\text{Length}] = [\text{Time}] = [\text{Energy}]^{-1}$
 To see this use $c = \frac{\text{Length}}{\text{Time}}$, now since velocity is dimensionless $\implies$ Length and Time have same dimensions.
 Use $\Delta E \Delta t \sim \hbar$, since $\hbar$ is dimensionless Energy will have dimension inverse of time.
- Consider the following table:

Quantity	SI	NU	Unit
Energy	ML^2T^{-2}	1	GeV
Momentum	MLT^{-1}	1	GeV
Time	T	-1	GeV^{-1}
Length	L	-1	GeV^{-1}
Cross Section(σ)	L^2	-2	GeV^{-2}
Action	ML^2T^{-1}	0	dimensionless
Electric Charge	IT	0	dimensionless

Here NU: Natural Units

Cross section is conventionally measured in barns $1 \text{ barn} = 1 \times 10^{-28} m^2$ and need convert accordingly.

- To interpret the result calculated using NU we need to convert it to SI units.
- insert $\hbar$ and c untill the dimensions match back
- $\hbar = 1.05457 \times 10^{-34} \text{ Js}$
 $c = 2.99 \times 10^8 \text{ m/s}$
 $\hbar c = 3.1615 \times 10^{-26} \text{ Js}$
 $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$
 $1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}$
 $1 \text{ GeV} = 1.602 \times 10^{-10} \text{ J}$

- Convert J to MeV

$$\begin{aligned}\hbar c &\sim \frac{3.1615 \times 10^{-26}}{1.602 \times 10^{-13}} \text{ MeV m} \\ &\sim 1.97 \times 10^{-13} \\ \hbar c &\approx 197.3 \text{ MeV fm} \quad (\text{fm: femtometer})\end{aligned}$$

Now in natural units $\hbar c = 1$, and hence

$$1 \text{ fm} = \frac{1}{197.3} \text{ MeV}^{-1} = 5.06 \text{ GeV}^{-1} \quad (1.1)$$

- Convert time s to GeV

$$\begin{aligned}\hbar &= \frac{1.05457 \times 10^{-34}}{1.602 \times 10^{-13}} \text{ GeV s} \\ &= \end{aligned}$$

Lecture 4

2.1 Special Relativity

- Laws of physics are same in all inertial reference frames
- the speed of light is constant in vacuum in all inertial frames

$$t' = \gamma(t - \beta x) \quad x' = \gamma(x - \beta t)$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \leftarrow \text{Lorentz Factor}$$

Length Contraction

$$L_0 = \gamma L$$

$$L_0 = x'_B - x'_A \quad L = x_B - x_A$$

★ Proton at LHC has $\gamma \sim 7000$. so the detector appears flattened by a factor of 7000

Time-Dilation

$$\Delta t = \gamma \Delta \tau$$

$$\gamma > 1 \quad \Delta t > \Delta \tau$$

Time dilation increases the lifetime of the particles

Experimental Evidence: Cosmic Rays

Half life of cosmic rays is $\tau_\mu = 2.2\mu\text{s}$. Therefore they can travel the distance $d = c\tau_\mu \sim 660\text{m}$. Hence should decay in atmosphere itself about 10 km high from surface.

But μ^- were observed at surface of earth so lifetime should be more.

2.2 4-Vector Notation

$$x^\mu = (x_0, x_1, x_2, x_3) = (t, x, y, z) = (t, \vec{x})$$

$$x_\mu = (t, -\vec{x})$$

$$x_\mu = \eta_{\mu\nu} x^\nu$$

$$\eta^{\mathbf{1}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

- Invariant Quantity: $a.b = a^\mu b_\mu = a_\mu b^\mu$
- Invariant Interval in space-time

$$\begin{aligned} ds^2 &= dx^\mu dx_\mu \\ &= dt^2 - dx_1^2 - dx_2^2 - dx_3^2 \\ &= dt^2 - d\vec{x}.d\vec{x} \end{aligned}$$

- the sign of ds^2 determines the causal structure

Time-like	Space-like	Light-like
$ds^2 > 0$	$ds^2 < 0$	$ds^2 = 0$
Causal connection is possible	No causal connection possible	Connection through light rays
There exists a frame in which events occur at same place	There exists a frame where events are simultaneous in time	Events lie on light cone

Proper Time

$$\begin{aligned} d\tau^2 &= dt^2 - d\vec{x}^2 \\ \Rightarrow d\tau &= dt \sqrt{1 - \beta^2} \\ dt &= \gamma d\tau \end{aligned}$$

Four-Velocity

$$\begin{aligned} U^\mu &= \frac{dx^\mu}{d\tau} \\ U^\mu &= \frac{dx^\mu}{dt} \frac{dt}{d\tau} = \gamma \frac{dx^\mu}{dt} = \gamma(1, \beta) \end{aligned}$$

U^μ is a normalized four-vector

$$U^\mu U_\mu = 1 \quad (2.1)$$

¹This is particle physics convention also called West-coast convention as people of west USA use this

Four-Momentum

$$p^\mu = mU^\mu = (m\gamma, m\gamma\vec{\beta})$$

$$E = m\gamma = \frac{m}{\sqrt{1-\beta^2}}$$

Mass Shell Condition

$$p^2 = p^\mu p_\mu = m^2 \quad (2.2)$$

$$\implies E^2 - \vec{p}^2 = m^2$$

$$E = \sqrt{|\vec{p}|^2 + m^2}$$

For massless particles: $p^2 = 0 \implies E = |\vec{p}|$ and $\beta = 1$

2.3 Decay Lifetimes

$$t_{\text{lab}} = \gamma\tau = \frac{E}{m}\tau$$

life-time of an unstable particle (τ) is defined in its rest frame

Pion-decay: $\pi^+ \rightarrow \mu^+ + \nu_\mu$ (two body decay)

mass of pion: 140 MeV

$$\tau_{\pi^+} = 2.6 \times 10^{-8} \text{s}$$